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THERAPEUTIC HOTLINE: SHORT PAPERS

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The efficacy of intradermal injection of type A botulinum toxin for facial rejuvenation

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Abstracts

The aim of this study is to evaluate the efficacy for effectiveness of type A botulinum toxin intradermal injection for facial rejuvenation. Forty female subjects were randomly divided into two groups: BoNTA group and control group. In BoNTA group, each subject's facial skin was treated with intradermal injection of BoNTA, and subjects of the control group were treated with intradermal saline solution injection. Subjects receiving one session of treatment and evaluations were conducted at baseline, four weeks, and twelve weeks after treatment. The outcome assessments included subjective satisfaction scale; blinded clinical assessment; and the biophysical parameters of roughness, elasticity, skin hydration, transepidermal water loss (TEWL), erythema, and melanin index. BoNTA group showed higher physician's global assessment score, subject satisfaction score, roughness, skin hydration, skin elasticity, and lower TEWL compared to that of control group at 12 weeks post-treatment. No significant difference was found among erythema and melanin index at baseline, four, and twelve weeks after treatment among the two major groups. In conclusion, intradermal BoNTA injection can be considered as an effective method for facial rejuvenation.

KEYWORDS

BoNTA, facial rejuvenation, intradermal injection

1 | INTRODUCTION

Botulinum Toxin Type A (BoNTA) is a powerful neurotoxin, that inhibits the release of several neurotransmitters, especially acetylcholine, and causes functional denervation of striated muscle postinjection (Jagdeo, Carruthers, & Smith, 2015). In the past 20 years, intramuscular injection with BoNTA as a cosmetic procedure has become more and more popular in facial rejuvenation field (Carruthers, Carruthers, Monheit, Davis, & Tardie, 2010; Hoenig & Hoenig, 2013; Raspaldo, Baspeyras, et al., 2011; Raspaldo, Niforos et al., 2011). In 2008, Chang et al. reported that the wrinkle-soothing effect on the middle and lower face can be achieved by intradermal injection of BoNTA (Chang et al., 2008). However, in their findings, no significant face-lifting effect of intradermal BoNTA injection was observed, and wrinkle-soothing effects by BoNTA injection contributing to the promotion of dermal collagen synthesis remain questionable. Our previous *in vitro* study through decreasing senescence-related proteins confirmed the direct anti-photoaging potential of BoNTA in ultraviolet B-induced premature senescence of human dermal fibroblasts (Permatasari, Hu, Zhang, Zhou, & Luo, 2014). However, BoNTA showing its direct skin rejuvenation effects have not been studied *in vivo* yet. This report illustrates the facial rejuvenating effect of intradermal injection of BoNTA in 20 Chinese women.

2 | SUBJECTS AND METHODS

2.1 | Subjects

Forty female subjects with Fitzpatrick photo types III and IV were enrolled in the study through an informed consent, and the Institutional Review Board of the First Affiliated Hospital of Nanjing Medical University approved this study. Patients included had an age range of 33–67 (mean age 40.5). Exclusion criteria are listed in Table 1.

2.2 | Methods

Subjects were randomly divided into two groups: 20 subjects received BoNTA intradermal injection therapy (BoNTA group) while another 20 subjects received saline solution intradermally (control group). Before treatment, a topical anesthetic cream (2.5% lidocaine and 2.5%

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TABLE 1 Exclusion criteria of subjects and skin measurements by biophysical evaluation methods taken at baseline, 4 weeks, and 12 weeks after treatment on both sides. ($\bar{x} \pm S$)

Groups	Baseline	4 weeks post-injection	12 weeks post-injection	Exclusion criteria
Erythema index				(1) Keloids and hypertrophic scars, cancer lesions, warts or skin infections in the area to be treated;
BoNTA group	292.32 ± 78.80	297.12 ± 84.25	264.12 ± 65.12	
Control group	290.14 ± 69.00	272.64 ± 45.22	285.56 ± 54.92	
Melanin index				(2) Viral herpes infections during the previous 6 months;
BoNTA group	202.76 ± 33.74	203.20 ± 31.66	193.50 ± 28.72	
Control group	192.76 ± 36.14	197.14 ± 24.75	190.60 ± 29.34	
TEWL				(3) Collagen disease, and autoimmune disease;
BoNTA group	22.03 ± 5.48	19.52 ± 3.54	16.56 ± 3.82*,**	
Control group	21.96 ± 4.77	21.66 ± 4.84	21.25 ± 3.45	
Hydration				(4) Individuals who had taken systemic isotretinoin, BoNTA injection or any facial rejuvenation procedures during the previous 12 months;
BoNTA group	56.04 ± 8.59	60.18 ± 6.72	67.73 ± 3.28*,**	
Control group	58.62 ± 7.02	55.84 ± 9.42	58.78 ± 5.24	
Elasticity				(5) Individuals who had used nonreabsorbable fillers;
BoNTA group	0.46 ± 0.12	0.48 ± 0.19	0.71 ± 0.07*,**	(6) Individuals who were undergoing treatment with anti-neoplastics, corticosteroids, or anticoagulants;
Control group	0.44 ± 0.10	0.45 ± 0.09	0.50 ± 0.08	
Skin surface roughness				(7) Individuals who were diabetic, pregnant, or breastfeeding
BoNTA group	8.60 ± 1.32	9.25 ± 1.42	10.88 ± 1.09*,**	
Control group	8.53 ± 1.02	8.84 ± 0.85	8.93 ± 1.56	

Note: *$p < .05$ compared with control side at same time point; **$p < .05$ compared with baseline.

prilocaine; Tsinghua Ziguang Co., Beijing, China) was applied on the skin 30 min before treatment and then completely removed. The BoNTA powder (HENGLI® Lanzhou Institute of Biological Products Co. Ltd., Lanzhou, Gansu, China) was dissolved into injectable saline solution and injected into the facial skin of each person with a multi-needle dermal injector system (Derma Shine, Panace, Inc., Seoul, Korea). A total 30 U BoNTA (dilluted in 2.6 milliliter saline solution, concentration at 12.5 U/ml) was injected into each subjects facial skin in BoNTA group, while 2.6 milliliter saline solution was injected into each subjects facial skin in control group. The injected depth was 0.8 mm in upper face, and 1.0 mm in middle and lower face, with an interval of 2 mm, having a volume of 2 μl in every point.

Measurements were conducted at baseline, four, and twelve weeks after the treatment. The outcome assessments included the subjective satisfaction scale (rated their improvement on a scale from 0 (aggravated) to 4 (much improved)), improvement score according to two blinded investigators (using standardized photography) using a well-established grading scale of 0 = < 25% (minimal), 1 = 26–50% (fair), 2 = 51–75% (good), 3 = 75–90% (excellent), 4 = 91–100% (clear) improvement, and biophysical measurements. The parameters assessed included erythema, melanin, transepidermal water loss (TEWL), elasticity, skin surface roughness, hydration, which were measured with respective probes (Courage & Khazaka electronic GmbH, Cologne, Germany). All measurements were performed on the both left and right cheeks of each subject.

2.3 | Statistical analysis

The results were analyzed with the paired *t* test using SPSS 15.0 software (SPSS, Inc., Chicago, IL). $p < .05$ was considered to be significant.

3 | RESULTS

Figure 1 shows representative clinical manifestations of both groups taken at indicated time points. During the evaluation periods, two blinded evaluators' scores increased successively in both groups. Twelve weeks after the treatment, BoNTA group clinical assessment score values were 2.30 ± 0.40 which was significantly higher than that of in control group (0.70 ± 0.60) ($p < .05$) (Figure 2a). During the evaluation period, the subjective satisfaction score in both groups had an obvious increment. Twelve weeks after the treatment, BoNTA group subjective satisfaction score values were 2.40 ± 0.50 which was significantly higher than that of the control group (0.75 ± 0.47) ($p < .05$) (Figure 2b).

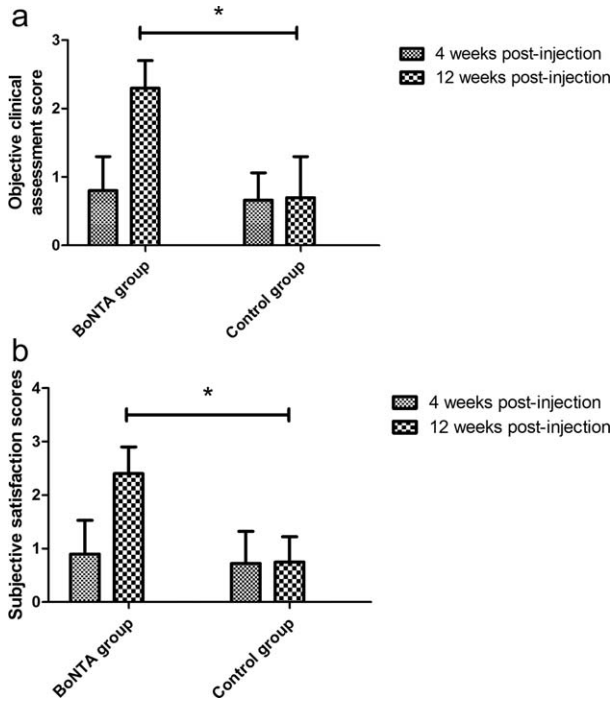


FIGURE 1 Subjective satisfaction scale and objective clinical assessment of outcome results in BoNTA and control groups. In BoNTA group, each subject's facial skin was treated with intradermal injection of BoNTA, and the control group subjects were treated with intradermal saline solution injection. Subjects received one session of treatment and evaluations were done as described in the Subjects and Methods section. Objective clinical assessment of outcome results (A) and subjective satisfaction scale (B) were taken four and twelve weeks after treatment.*$p < .05$ with the paired *t* test

As shown in Table 1, there were no significant differences in the facial indexes between the two groups before injection. According to statistical analysis, there were no significant differences among two groups in Erythema and Melanin index at four and twelve weeks after the treatment, respectively. At twelve weeks after the treatment, indexes of TEWL, elasticity (R2), skin surface roughness, hydration of the BoNTA group showed improvement compared to the control group. Compared with baseline, the facial indexes showed no significant differences among different time points after injection in control group. The indexes of TEWL, elasticity (R2), skin surface roughness, and hydration of the BoNTA group at twelve weeks after the treatment showed significant improvements than that in baseline respectively.

4 | DISCUSSION

As to erythema and melanin index, BoNTA group showed no statistical differences with control group up to 12 weeks after injection. Although BoNTA can be applied for the treatment of facial flushing (Lin, Wang, Fuh, & Chen, 2013), it did not induce excessive angiogenesis as shown in the present study. Regarding an increase in the level of TEWL and a decrease in the hydration content in the stratum corneum, 12 weeks after the treatment in BoNTA group, these results suggested that BoNTA can promptly improve skin barrier function. The reduction in TEWL may be caused at least in part by the known effect of BoNTA products to reduce sweating (Dressler, 2010). It is not known whether there is any additional mechanism of action by which BoNTA has an effect on TEWL. Increased skin hydration may improve skin quality. It has been reported that intradermal injection of BoNTA can cause a reduction in sebum production (Rose and Goldberg, 2013).

It was confirmed by biophysical analysis that the skin elasticity and skin surface roughness in BoNTA group were better than control group 12 weeks post-treatment. These results suggest that intradermal injection of BoNTA have facial rejuvenation effect. We speculated that this effect was mainly due to increment of dermal collagen synthesis, but not because of the blockage of BoNTA on the muscle fiber due to the following two reasons. First, Oh et al and our *in vitro* study confirmed that BoNTA had a significant effect to increase collagen production and down-regulate its degradation in dermal fibroblasts (Oh et al., 2012; Permatasari et al., 2014). *In vivo* study, more collagen deposition was observed by histological examination 2 months after BoNTA injection (Chang et al., 2008). Second, the facial rejuvenation effect was subjectively stated and objectively recorded at 12 weeks after BoNTA injection, and no significant improvement was achieved at earlier time points. From previous studies of nonablative modalities such as 1320 nm Nd:YAG laser and the radiofrequency-based dermal remodeling device, clinical assessable improvement from collagen synthesis might appear after 2–3 months (Nouri, Rivas, Bouzari, & Faghih, 2006; Zelickson et al., 2004). While low-dose intramuscular BoNTA injection

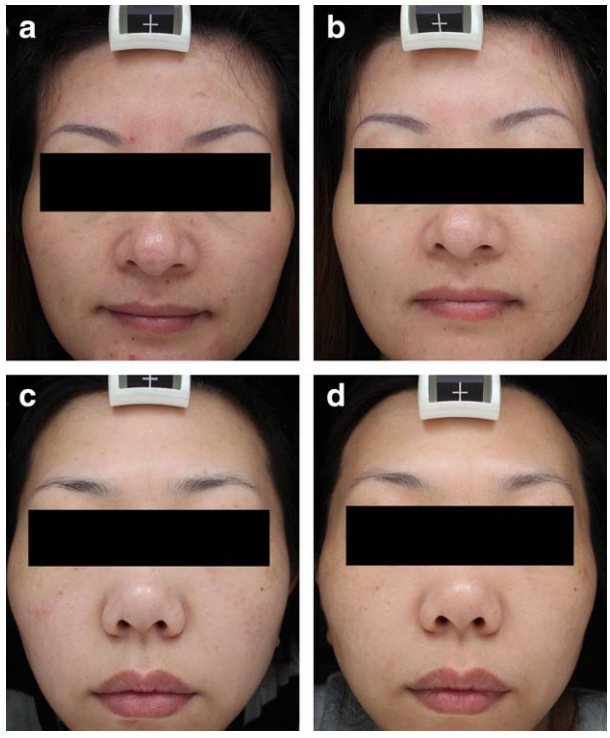


FIGURE 2 Representative clinical photographs. (a, c) Baseline, (b) 12 weeks after the intradermal BoNTA injection, (d) 12 weeks after the intradermal saline solution injection

showed its rejuvenation effects by muscle relaxation appeared within 4 weeks after injection, and gradually diminished 2–3 months later (Dayan, Arkins, Patel, & Gal, 2010; Yu, Chen, & Chen, 2007). The time course of the facial rejuvenation effect was more parallel with that of the dermal collagen remodeling way.

One major limitation of this study is that it is merely a small sample preliminary clinical study. A larger sample randomized controlled clinical trial with longer observation periods need to verify the results of this study. Besides that, further studies are also needed to compare the facial rejuvenation efficacy between intradermal injection of BoNTA and conventional intramuscular BoNTA injection. Conclusively relying on the results of present data, we suggest intradermal BoNTA injection as an effective method for facial rejuvenation.

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CONFLICT OF INTEREST

The authors report no conflicts of interest.

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